PROBLEM 24

Particle on a Circumference with a Spring

Original statement · Problems.pdf, p. 12

Wire & Elastic Potential

A heavy particle P is constrained to move along a circumference and it is attached to a spring as shown in the figure. The mass of the particle is m, and the radius of the circumference is a. The other end of the spring is attached to a point of the y axis with coordinates $(0, \beta a)$ and the constant of the spring is equal to k.

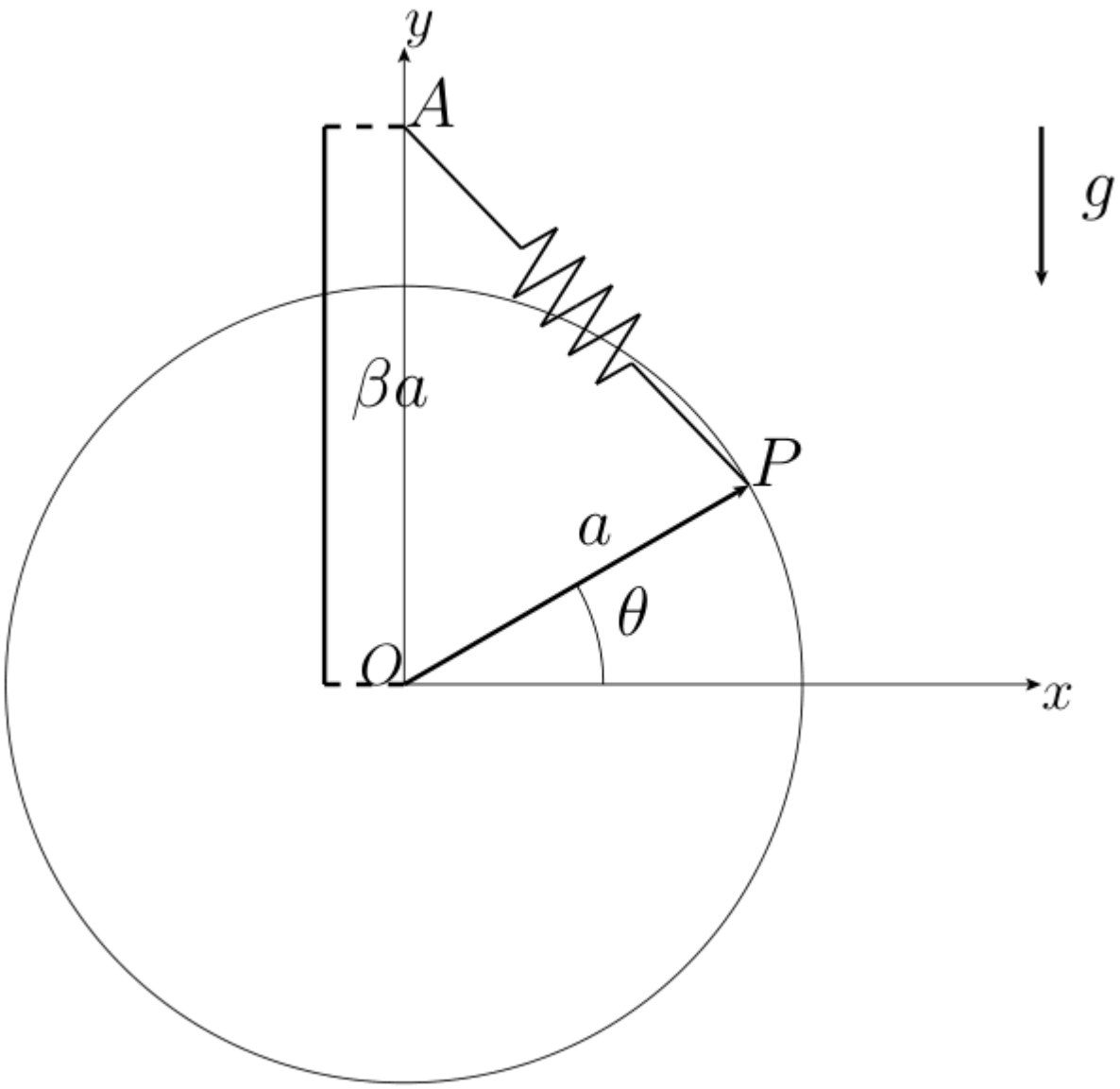


Figure: Particle P on the circumference of radius a centred at O, spring from A = (0, βa) to P, angle θ from the x axis (Problems.pdf, p. 12).

(a) Compute the potential of the forces acting on the particle as a function of the angle θ (shown in the figure).

(b) Identify the equilibrium positions.

The particle departs from the point $(a, 0)$ of the circumference with an initial velocity $\mathbf{v}_0 = v_0\mathbf{j}$.

(c) Reduce the motion to integrals (i.e., quadratures).

(d) If $\beta = \frac{mg}{ka}$, obtain the position of the particle as a function of time.

Consider now the situation in which an additional force acts on the particle. The force is proportional to, and in the same direction as, the instantaneous velocity. The constant of proportionality is $2m\mu$. The initial conditions are the same as before.

(e) Obtain the position of the particle as a function of time.

(f) Obtain the reaction of the circumference.

Worked Solution (4-Phase Methodology)

PHASE 1

Constraints, Degrees of Freedom & Coordinates

- **System:** a heavy particle P of mass m slides on a vertical circular wire of radius a centred at O. A linear spring of stiffness k and zero natural length joins P to the anchor $A = (0, \beta a)$ on the y-axis. Gravity acts along $-\mathbf{j}$.
- **Constraints:** P stays on the circle $x^2 + y^2 = a^2$ in the plane Oxy (holonomic, scleronomic, smooth, bilateral). The wire gives a reaction normal to it, with an in-plane component and, if needed, a component perpendicular to the plane (zero here). $\mathrm{CDOF} = 2 - 1 = 1$, with the angle θ measured counterclockwise from the positive x-axis.
- **Initial state:** P starts at $(a, 0)$, i.e. $\theta(0) = 0$, with $\mathbf{v}_0 = v_0\,\mathbf{j}$, i.e. $\dot\theta(0) = v_0/a$.
- **Parameters:** $m, a, k, g, \beta, v_0 > 0$. Parts (e) and (f) use $\beta = mg/(ka)$ (the case of part (d)), and the additional force has the constant $2m\mu$.

PHASE 2

Reference Frames & Vector Bases

- **Inertial frame:** $\mathcal{F}_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ with $\mathbf{j}$ vertical upward.
- **Polar basis:** $\mathbf{e}_r = \cos\theta\,\mathbf{i} + \sin\theta\,\mathbf{j}$ and $\mathbf{e}_\theta = -\sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j}$, a rotation of $\mathcal{B}_0$ by θ about $\mathbf{k}$ (matrix $\begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $\det = \cos^2\theta + \sin^2\theta = 1$; RR^T has diagonal entries $\cos^2\theta + \sin^2\theta = 1$ (and 1) and off-diagonal entries $\cos\theta\sin\theta - \sin\theta\cos\theta = 0$, so $RR^T = I$), with $\dot{\mathbf{e}}_r = \dot\theta\,\mathbf{e}_\theta$ and $\dot{\mathbf{e}}_\theta = -\dot\theta\,\mathbf{e}_r$ (Poisson).

PHASE 3

Derivation

3.1 Kinematics

Pedagogical Justification: the position of P and the anchor give the spring vector, and the polar formulas give the velocity and acceleration:

$$\mathbf{r}_P = a\,\mathbf{e}_r = a\cos\theta\,\mathbf{i} + a\sin\theta\,\mathbf{j}, \qquad \mathbf{v} = a\dot\theta\,\mathbf{e}_\theta, \qquad \mathbf{a} = -a\dot\theta^2\,\mathbf{e}_r + a\ddot\theta\,\mathbf{e}_\theta$$

$$\overrightarrow{AP} = \mathbf{r}_P - \beta a\,\mathbf{j} = a\cos\theta\,\mathbf{i} + a(\sin\theta - \beta)\,\mathbf{j}, \qquad \left|\overrightarrow{AP}\right|^2 = a^2\left(1 + \beta^2 - 2\beta\sin\theta\right)$$

3.2 Forces

- **Gravity:** $-mg\,\mathbf{j}$, with $\mathbf{F}_g \cdot \mathbf{e}_r = -mg\sin\theta$ and $\mathbf{F}_g \cdot \mathbf{e}_\theta = -mg\cos\theta$.
- **Spring:** $\mathbf{F}_s = -k\,\overrightarrow{AP}$, with $\mathbf{F}_s \cdot \mathbf{e}_r = -ka\left(1 - \beta\sin\theta\right)$ and $\mathbf{F}_s \cdot \mathbf{e}_\theta = ka\beta\cos\theta$.
- **Reaction:** $\mathbf{N} = N\,\mathbf{e}_r$ (normal to the wire; $N > 0$ is outward). **Additional force** (parts e, f): $\mathbf{F}_d = 2m\mu\,\mathbf{v}$, along the velocity, as the statement says. **Inertia forces:** none.
- **Potential.** Gravity and the linear spring are conservative, so the total potential energy is the sum of $V_g = mgy = mga\sin\theta$ and $V_s = \frac{1}{2}k\left|\overrightarrow{AP}\right|^2$ (the work of the wire reaction is zero):

$$V(\theta) = mga\sin\theta + \tfrac{1}{2}ka^2\left(1+\beta^2-2\beta\sin\theta\right) = \tfrac{1}{2}ka^2\left(1+\beta^2\right) + (mg - k\beta a)\,a\sin\theta$$

3.3 Equations of Motion

- **Tangential equation** (no reaction): $ma\ddot{\theta} = \mathbf{F}_g\cdot\mathbf{e}_\theta + \mathbf{F}_s\cdot\mathbf{e}_\theta + \mathbf{F}_d\cdot\mathbf{e}_\theta = -(mg - k\beta a)\cos\theta + 2m\mu a\dot{\theta}$. For $\mu = 0$ this is $ma\ddot{\theta} = -\frac{1}{a}\frac{dV}{d\theta}$, i.e. the equation of motion of the conservative problem.
- **Radial equation** (gives the reaction): $-ma\dot{\theta}^2 = N - mg\sin\theta - ka\left(1-\beta\sin\theta\right)$.
- Two equations for the two unknowns $\theta(t)$ and $N(t)$. The reaction is decoupled and is computed after $\theta(t)$.
- **Energy (conservative case $\mu = 0$):** the reaction does no work, so $E = \frac{1}{2}ma^2\dot{\theta}^2 + V(\theta)$ is a first integral.

3.4 Integration and Answers

(a) Potential of the forces

$$\boxed{V(\theta) = \tfrac{1}{2}ka^2\left(1+\beta^2\right) + (mg - k\beta a)\,a\sin\theta}$$

(b) Equilibrium positions

Pedagogical Justification: equilibrium requires $\dfrac{dV}{d\theta} = 0$:

$$\frac{dV}{d\theta} = (mg - k\beta a)\,a\cos\theta = 0$$

- If $mg - k\beta a \neq 0$: $\cos\theta = 0$, i.e. $\theta_1 = \pi/2$ (top) and $\theta_2 = -\pi/2$ (bottom). Since $V'' = -(mg - k\beta a)a\sin\theta$, the top is a minimum (stable) if $mg < k\beta a$ and the bottom is a minimum if $mg > k\beta a$.
- If $\beta = \dfrac{mg}{ka}$: $dV/d\theta \equiv 0$, so **every** position is an equilibrium (neutral equilibrium).

(c) Reduction to quadratures

Pedagogical Justification: the energy is constant for $\mu = 0$. At $t = 0$, $V(0) = \frac{1}{2}ka^2(1 + \beta^2)$ and $\dot{\theta}(0) = v_0/a$, so

$$\tfrac{1}{2}ma^2\dot{\theta}^2 + V(\theta) = \tfrac{1}{2}mv_0^2 + V(0) \implies \dot{\theta} = \pm\sqrt{\frac{v_0^2}{a^2} - \frac{2\,(mg - k\beta a)}{ma}\sin\theta}$$

$$\int_0^{\theta(t)} \frac{d\theta'}{\sqrt{\dfrac{v_0^2}{a^2} - \dfrac{2\,(mg - k\beta a)}{ma}\sin\theta'}} = \pm\int_0^t dt' = \pm t$$

(d) Case $\beta = mg/(ka)$

Pedagogical Justification: the coefficient $mg - k\beta a$ vanishes, so V is constant and the tangential force is zero: $ma\ddot{\theta} = 0$. Integrating with Barrow's rule:

$$\dot{\theta}(t) = \dot{\theta}(0) = \frac{v_0}{a}, \qquad \theta(t) = \int_0^t \frac{v_0}{a}\,dt' = \left[\frac{v_0 t'}{a}\right]_0^t = \boxed{\frac{v_0}{a}\,t}$$

The particle circles the wire at constant angular speed.

(e) Position with the additional force along the velocity

Pedagogical Justification: with $\beta = mg/(ka)$ the gravity and spring contributions to the tangential equation cancel, so only the additional force remains: $ma\ddot{\theta} = 2m\mu\,(a\dot{\theta})$, i.e.

$$\ddot{\theta} - 2\mu\dot{\theta} = 0$$

- Let $\Omega(t) = \dot{\theta}$; then $\dfrac{d\Omega}{\Omega} = 2\mu\,dt$, and integrating from $\Omega(0) = v_0/a$ (Barrow):

$$\int_{v_0/a}^{\Omega} \frac{d\Omega'}{\Omega'} = \int_0^t 2\mu\, dt' \implies \left[\ln \Omega'\right]_{v_0/a}^{\Omega} = \ln \frac{a\Omega}{v_0} = 2\mu t \implies \frac{a\Omega}{v_0} = e^{2\mu t} \implies \Omega(t) = \frac{v_0}{a} e^{2\mu t}$$

- With $w = 2\mu t'$, $dw = 2\mu\, dt'$, the antiderivative is $\int e^{2\mu t'} dt' = \frac{1}{2\mu}\int e^{w}\, dw = \frac{e^{2\mu t'}}{2\mu}$, and with $\theta(0) = 0$:

$$\theta(t) = \frac{v_0}{a}\int_0^t e^{2\mu t'} dt' = \frac{v_0}{a}\left[\frac{e^{2\mu t'}}{2\mu}\right]_0^t = \frac{v_0}{a}\cdot\frac{e^{2\mu t} - 1}{2\mu} = \boxed{\frac{v_0}{2\mu a}\left(e^{2\mu t} - 1\right)}$$

The force acts along the velocity, so it feeds energy to the particle and the angular speed grows exponentially.

> **Reading of the statement**
> The statement says that the additional force is proportional to, and in the **same direction as**, the instantaneous velocity (checked on the PDF page), so it is $+2m\mu\,\mathbf{v}$ and the solution above follows it literally. If the force were meant to oppose the velocity (viscous damping), the sign changes: $\ddot\theta + 2\mu\dot\theta = 0, \theta(t) = \dfrac{v_0}{2\mu a}\left(1 - e^{-2\mu t}\right)$, which tends to $\theta_\infty = v_0/(2\mu a)$ at rest, and the reaction below becomes $N = ka - \dfrac{mv_0^2}{a}e^{-4\mu t}$.

(f) Reaction of the circumference

> **Pedagogical Justification:** the radial equation of 3.3, with $\mathbf{F}_g\cdot\mathbf{e}_r$ and $\mathbf{F}_s\cdot\mathbf{e}_r$ of 3.2 (the additional force is tangential, so it has no radial component):

$$N(t) = -ma\dot\theta^2 + mg\sin\theta + ka\left(1 - \beta\sin\theta\right)$$

- With $\beta = mg/(ka)$ the terms in $\sin\theta$ cancel ($mg\sin\theta - ka\beta\sin\theta = 0$), and $\dot\theta = (v_0/a)e^{2\mu t}$:

$$\boxed{N(t) = ka - \frac{mv_0^2}{a}\, e^{4\mu t}}$$

N is the outward component. It starts at $ka - mv_0^2/a$ and becomes more and more negative: the wire must push the particle inward with a force that grows as $e^{4\mu t}$.

PHASE 4

Interpretation & Checks

- **Dimensional check:** $[ka] = \mathrm{N}$ and $[mv_0^2/a] = \mathrm{N}$; μ has units s^{-1}, so $2\mu t$ is dimensionless and $[v_0/(2\mu a)] = 1$ (rad).
- **Limits:** for $\mu \to 0$, $\theta \to (v_0/a)t$, the result of (d), and $N \to ka - mv_0^2/a$, the preload of the spring minus the centripetal force, constant.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 24 in its solution keys, so the results are verified by independent derivation. Sympy confirms the radial equation $N = -ma\dot{\theta}^2 + mg\sin\theta + ka(1 - \beta\sin\theta)$ for either sign of the additional force and the reduction to $N = ka - (mv_0^2/a)e^{\pm 4\mu t}$ when $\beta = mg/(ka)$. Numerical integration of $\ddot{\theta} = 2\mu\dot{\theta}$ ($a = 1$, $v_0 = 1.7$, $\mu = 0.3$) gives $\theta(2) = 6.57366$, equal to the closed form; the damping reading gives 1.97995, also equal to its closed form.

Source: Problems.pdf